BarSync

MIDI CLOCK BAR COUNTER & VISUALIZER

Quickstart Guide

Always in Time.

Firmware 1.3.0

OVERVIEW

What is BarSync?

BarSync keeps you on top of your track's arrangement — live or in the studio. It receives your sequencer's MIDI clock and counts and visualizes the bars elapsed since a starting point you define yourself — either the first MIDI clock signal, or a manual reset via button. That way you always know exactly which bar you're in, and drops, breaks and builds never catch you off guard.

Features

- Bar/beat counter, synced to the incoming MIDI clock (24 PPQN)
- Grid progress bar (x1 to x128 bars) — footprint stays constant, tile size adapts automatically
- Two grid display modes: CYCLE (repeating cycle) or SCROLL (continuous, with markers scrolling into view)
- Marker Editor: place your own markers on the bar grid (4 symbols) — for arrangement points, phrase starts, or band cues
- End Bar: automatically loop, stop, or continue the count at a defined bar — fixed, or at your last placed marker
- 5 selectable time signatures (3/4, 4/4, 5/4, 6/8, 7/8)
- Reset button with two stages (bar end / cycle end), plus a standalone SET 1.1 function to instantly lay down a new beatgrid (quantized or instant)
- MIDI Monitor built into the device (Note/CC/PC/Pitch-Bend log, MIDI clock and RUN/STOP graph)
- Nudge mode to manually compensate for clock phase drift
- MIDI Thru to pass the signal on to further devices
- Standby (light sleep) on MIDI inactivity
- Full settings menu right on the device — no app, no software needed

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GETTING STARTED

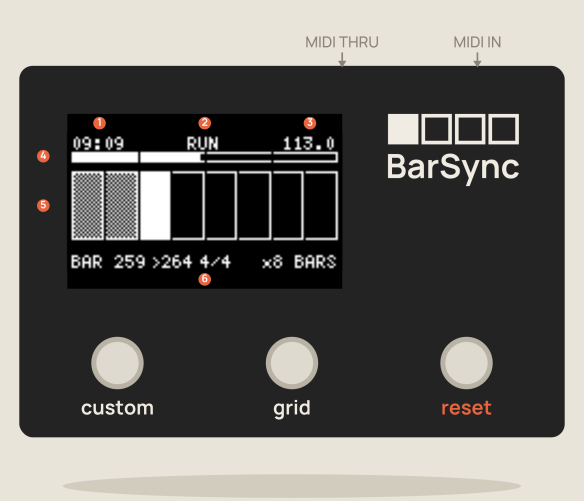
Connection & First Start

1. **Connect your MIDI source.**
   Connect your sequencer's/DAW interface's MIDI OUT to BarSync's **MIDI IN**.
2. **Optional: pass it on.**
   MIDI THRU lets you forward the incoming signal unchanged to another device — handy when BarSync sits in the middle of an existing MIDI chain.
3. **Power it up.**
   BarSync starts automatically and shows the retro-style boot screen for about 5 seconds.
4. **Start the MIDI clock.**
   As soon as your sequencer is running, the status switches from **STOP** to **RUN**, and the bar/beat counter starts running.
5. **Set a reference point (optional).**
   The 1.1 is set automatically as soon as BarSync receives your sequencer's **Start/Run signal** — you don't have to do anything for that. If you'd rather start counting from a specific point instead, just press the **Reset** button.

> If no MIDI clock arrives for a while, the display shows **"NO MIDI-CLOCK"** instead — a cue to check your MIDI cable and source (see Troubleshooting on the last page).

OPERATION

The Three Buttons



The Display at a Glance

NO.	SHOWS
①	Play time (mm:ss) since the last start/reset
②	Status — RUN or STOP
③	Tempo in BPM
④	Progress within the current bar
⑤	Progress within the current grid cycle — one tile = one bar
⑥	current bar, end of cycle, time signature, grid size

custom

- **Short press:** SET 1.1 — instantly lays down a new beatgrid (default assignment).
- **Hold 1 second:** toggle MIDI Monitor on/off.
- **Together with Grid:** toggle Nudge mode on/off.
- **Together with Reset (hold 1s):** opens OPERATION SETUP (grid mode, marker editor, end bar).

grid

- **Short press:** cycles the number of bars shown in the grid (x1 to x128).

reset

- **Short press (Reset 1):** waits for the end of the current bar.
- **Hold (Reset 2):** waits for the end of the current grid cycle.
- **Hold 1 second while booting:** opens the settings menu.
- **On the MIDI Monitor screen:** clears the note/CC list instead.

The Custom button is freely assignable — the functions shown here match the firmware default (Switches › Custom › Function).

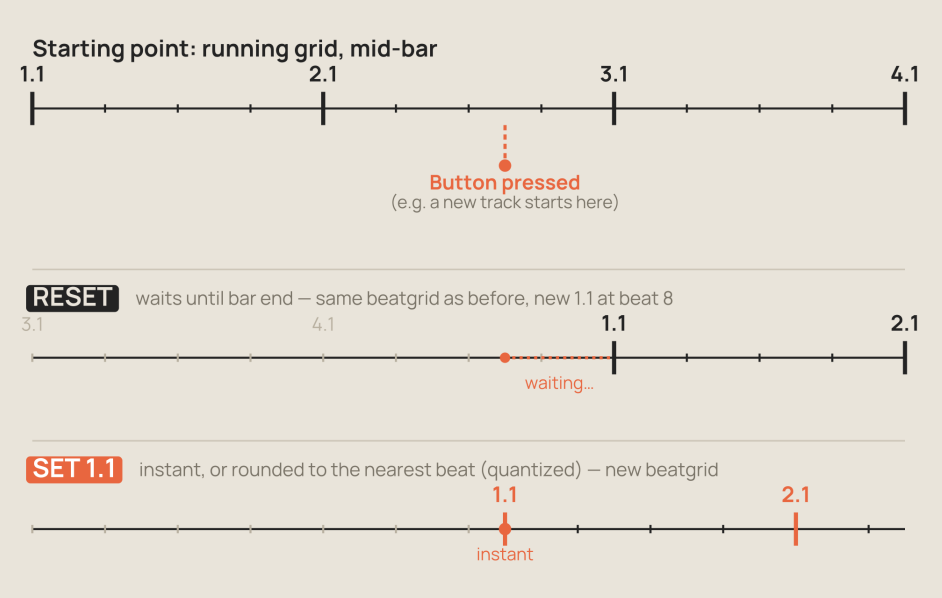
OPERATION IN DETAIL

Reset vs. SET 1.1 — What's the Difference?

The 1.1 is set all by itself to begin with: as soon as BarSync receives your sequencer's **Start/Run signal**, counting starts automatically at 1.1 — without you pressing any button.

You only need **SET 1.1** when you want to correct that automatically-set point afterwards — for example when you've started a new song during your live set and your track's 1.1 no longer "sits" on BarSync's 1.1. Sure, you could restart the sequencer and everything would line up again. But during a performance that's obviously something you'd rather avoid. That's where SET 1.1 comes in: pressing the **Custom** button lays down a new beatgrid with 1.1 exactly at that moment.

The two **Reset** stages (1 and 2) build directly on that beatgrid and simply reset the bar counter including the grid (and, depending on settings, the play time) — always in sync with the current beatgrid.


Starting point: running grid, mid-bar
1.1
2.1
3.1
4.1
Button pressed
(e.g. a new track starts here)
RESET
waits until bar end — same beatgrid as before, new 1.1 at beat 8
3.1
4.1
1.1
2.1
waiting...
SET 1.1
instant, or rounded to the nearest beat (quantized) — new beatgrid
1.1
2.1
instant

FUNCTION	EFFECT
Reset	Waits for the end of the current bar (Reset 1) or cycle (Reset 2) and then resets to 1.1. The **beatgrid stays unchanged** — the new bar boundaries land exactly where a bar boundary already was.
SET 1.1	Lays down a **completely new beatgrid** instantly (Instant) or rounded to the nearest beat (Quantized) — regardless of where the previous beatgrid sat. As a result, every following bar boundary can shift compared to before.

> Rule of thumb: if your set runs continuously within the same beatgrid, **Reset** is all you need. Only once a new song section starts with its own downbeat that doesn't match your current count do you need **SET 1.1** to re-anchor the beatgrid.

CONFIGURATION

Device Setup

Open it: hold the Reset button for 1 second while powering on.

SETUP
- TIMESIG — time signature, direct value (3/4, 4/4, 5/4, 6/8, 7/8), default 4/4
- SWITCHES ›
  - CUSTOM › Function (Timesig / SET 1.1) — depending on Function, underneath: time-signature selection or Mode (Quantized/Instant)
  - GRID › choice of which divisor steps are selectable
  - RESET › playtime reset per stage (Yes/No)
- DISPLAY ›
  - CONTRAST, INVERT (swap black/white), GRID LINES (separators every 32 bars, toggleable)
- STANDBY — on/off, sleep delay
- DEFAULTS — load factory settings (with confirmation)

Navigating the Menu

BUTTON	FUNCTION IN THE MENU
Custom	move up one line
Grid	move down one line
Reset (short)	open selection / change value
Reset (hold 1s)	back one level
Reset (on page 1, hold 3s)	save, exit menu, restart

CONFIGURATION

Operation Setup

Open/exit it: hold **Custom + Reset** together for 1 second — during normal operation, not while booting. Holding the same combo again exits the menu from any screen inside it and saves automatically.

OPERATION SETUP
- GRID MODE — CYCLE (repeating cycle) or SCROLL (continuous, markers scroll in from the right)
- SET MARKERS — opens the marker editor, see below
- END BAR — OFF / LAST MARKER / MANUAL
- END BAR ACT — LOOP / STOP / CONTINUE (only shown when END BAR ≠ OFF)

Navigating the Menu

BUTTON	FUNCTION IN THE MENU
Custom	move up one line
Grid	move down one line
Reset (short)	select the row / toggle its value (GRID MODE, END BAR, END BAR ACT directly; with END BAR = MANUAL a BAR NUMBER row also appears)
Reset (hold 1s)	back one level

Placing Markers (SET MARKERS)

The marker editor lets you tag individual bars with one of four symbols — e.g. for arrangement points, phrase starts, or band cues. It temporarily replaces the beat bar with two navigation arrows.

BUTTON	FUNCTION IN THE MARKER EDITOR
Custom	move the cursor back one bar (left arrow only shown when possible)
Grid	move the cursor forward one bar (right arrow only shown when possible)
Custom/Grid (hold >1s)	fast-page in steps of 16 bars; Custom and Grid are interlocked — only one direction at a time
Custom + Grid together	cycles the tool: square → circle → triangle → X → DEL (delete)
Reset (short)	places the marker with the current tool at the cursor — or erases it when the tool is DEL
Custom + Reset (hold 1s)	exits the editor, saves, and returns to OPERATION SETUP

Display at the bottom: current bar on the left (with a blinking cycle-end number), number of placed markers on the right (out of a max. of 500), the current tool, and a "RESET = PLACE" / "RESET = DELETE" hint in the middle.

DIAGNOSTICS

MIDI Monitor

The built-in MIDI Monitor shows you, right on the device, what's actually arriving on the MIDI line — handy for debugging your setup without any separate software or a computer sitting next to you.

Show/Hide

Hold the Custom button for 1 second (default assignment).

What's Shown?

- Event log of the last 6 Note On/Note Off, CC, Program Change and Pitch Bend messages, including channel — newest message on top
- MIDI clock graph: one spike per detected beat, the downbeat (1.1) drawn twice as wide as a regular beat
- MIDI RUN/STOP trace as a level graph, with a plain-text "RUN = HIGH" / "RUN = LOW" status

> On the Monitor screen, the **Reset** button takes on a different job: it clears the note/CC list instead of triggering the normal reset function.

> If no MIDI clock arrives for more than 5 seconds, "NO CLOCK" blinks in place of the clock graph instead.

REFERENCE

Troubleshooting & Quick Reference

PROBLEM	SOLUTION
Display stays dark	Check power supply; check contrast in the setup menu (Display › Contrast)
"NO MIDI-CLOCK" is shown	Check the MIDI cable and source; make sure the clock is actually running
Buttons don't respond / respond incorrectly	Restart the device; if in doubt, load factory settings (Setup › Defaults)
Device only responds after a delay following a long pause	Normal behavior: BarSync goes into standby (light sleep) on MIDI inactivity and wakes automatically on new activity

Quick Reference

BUTTON	FUNCTION
Custom (short)	SET 1.1 — lay down a new beatgrid*
Custom (hold 1s)	toggle MIDI Monitor on/off*
Grid (short)	cycle the number of bars shown (x1–x128)
Reset (short/hold)	Reset stage 1/2
Custom + Grid	toggle Nudge mode on/off*
Hold Reset while booting	settings menu

*The Custom button is freely assignable — this reflects the firmware default.

Full documentation, schematic, firmware and 3D-printable enclosure:
github.com/barna81/BarSync